

Block to Python Bridge

Project: Verified Study Buddy

1. Your mission

Your learning target

I can translate a short paper calculation into a Python script.

I can predict and explain the order of executed statements.

I can keep expected and observed output separate when checking a program.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Requests Responses and JSON. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Conversation State and Memory. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Model Memory vs Retrieval. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Week 23: Block-to-Python Bridge

Variables, conditions, lists, loops, and functions carry the same logic across representations.



CREATOR MISSION

Translate a block workflow into readable Python.

Creator name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

Four packs contain six counters each. What steps find and show the total?

This week's end goal

By the end, I can explain block to python bridge using a traced example and a stated limitation.

2. Notice the evidence

PICTURE STORY

Notice → Predict → Explain

Begin with visible evidence. Do not explain more than the picture and information support.



MYSTERY

The learners must use block-to-python bridge to solve a problem. What information do they have? What decision or calculation do they make? What would count as evidence that it worked?

Three observations (what you can point to):

Write or draw your reasoning and evidence.

My prediction and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

In Grade 6 you traced rules on paper. Now you will write a small Python program and check what it actually does. Begin with a counter-packing calculation, not the whole activity adviser. The requirement is to find the total counters in equal packs and display that total. For this first program, the supplied quantities are whole, nonnegative numbers; it does not yet validate typed input.

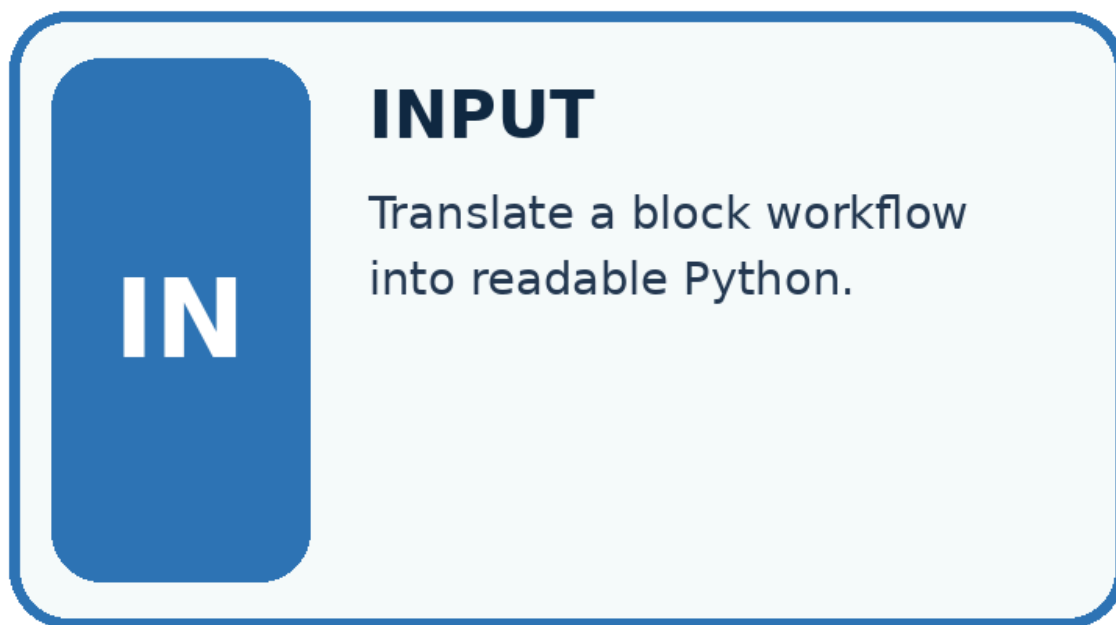
Use a teacher-prepared Python 3 editor that can run a saved .py file. Open a new file called counters.py, type the code exactly as shown in the next section, save it, and use the editor's Run command. Read the output area separately from the source editor. No account, AI tool, package installation or personal data is needed. If a device is unavailable, trace on paper and label the result predicted rather than executed.

3. See how it works

VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



↓



SYSTEM

Variables, conditions, lists, loops, and functions carry the same logic across representations.



OUT

EVIDENCE

Translate a block workflow into readable Python.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

A calculation from paper to output

Paper step · Python instruction · Effect

Paper step: Record pack count

Python instruction: `packs = 4`

Effect: Name holds 4

Paper step: Multiply two quantities

Python instruction: `total = packs * per_pack`

Effect: Name holds product

Paper step: Show the answer

Python instruction: `print(total)`

Effect: Displays the value

Assign `per_pack` before the multiplication. This diagram summarises the steps; the complete script below includes that assignment.

```
packs = 4

per_pack = 6

total = packs * per_pack

print(total)
```

Read the script from top to bottom. The equals sign assigns the value on its right to the name on its left. The star means multiplication. `print(total)` displays the value of `total`; it does not display the letters in the name. Type ordinary straight characters and keep the parentheses. The Python prompt symbols `>>>` are not part of this saved script.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

CASE GALLERY

Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.



NORMAL
Expected learner request



AMBIGUOUS
Two plausible meanings

NORMAL CASE: predict and name evidence. BOUNDARY CASE: predict and name evidence.



MISSING SOURCE

Evidence unavailable



ADVERSARIAL

Rules or trust challenged

MISSING INFORMATION: predict and name evidence. TRANSFER CASE: predict and name evidence.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change: Replace the last page count 90 with 100.

Prediction: The final count becomes 1.

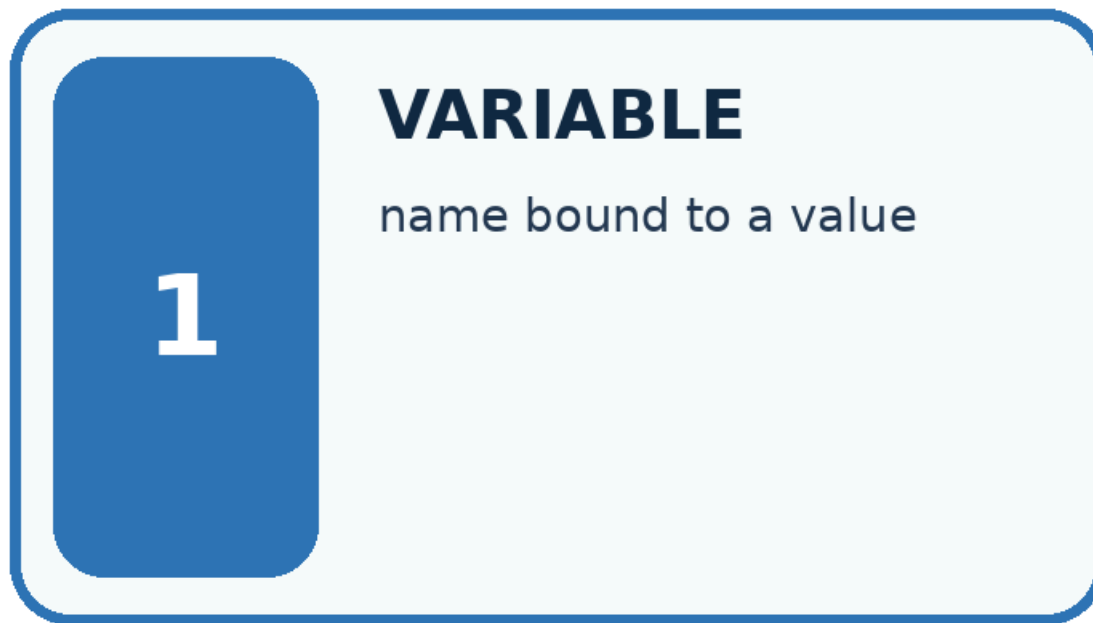
Reason: The comparison is strictly less than 100, so the boundary value does not qualify.

5. Words that unlock the idea

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.



2

LIST

ordered collection

3

FUNCTION

reusable named procedure

4

CONDITION

expression selecting a path

Word · My own example · Not this / boundary

Word: variable

My own example: A named reference to a value that can be used or changed in a program.

Not this / boundary: _____

Word: list

My own example: An ordered collection of items.

Not this / boundary: _____

Word: function

My own example: A named unit of code that performs a task, optionally receiving arguments and returning a value.

Not this / boundary: _____

Word: condition

My own example: A statement evaluated as true or false to help select an action.

Not this / boundary: _____

Words used in the closer explanation

Word · Meaning

Word: Algorithm

Meaning: An ordered procedure for solving a stated problem

Word: Program

Meaning: Instructions written in a language a computer can execute

Word: Interpreter

Meaning: Software that executes Python instructions

Word: Assignment

Meaning: Binding a name to the value of an expression

Word: Output

Meaning: Information a program displays or returns

6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

WEEK 23 • WORKED EVIDENCE

- 1 Block: for each passage, check relevance.
- 2 Python: for passage in passages: ...
- 3 Condition keeps passages with score $\geq$ threshold.
- 4 A function packages the selection logic and can be tested with normal and edge lists.

1. Block: for each passage, check relevance.
2. Python: for passage in passages: ...
3. Condition keeps passages with score $\geq$ threshold.
4. A function packages the selection logic and can be tested with normal and edge lists.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

After line · Names and values · Displayed output

After line: packs = 4

Names and values: packs is 4

Displayed output: Nothing yet

After line: `per_pack = 6`

Names and values: `packs` is 4; `per_pack` is 6

Displayed output: Nothing yet

After line: `total = packs * per_pack`

Names and values: `total` is 24

Displayed output: Nothing yet

After line: `print(total)`

Names and values: Values are unchanged

Displayed output: 24

Write 24 as the expected output before running. Then copy the actual output into a separate column. A matching result supports this case, not every possible input. A saved file may contain an old edit: save again and rerun the whole script. Do not type the expected number into the output record without running.

Why is there no displayed output after the third line? What changes if `print(total)` is placed first in a fresh run, before any assignment? Explain using which names exist.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Block to Python Bridge

Translate a “count qualifying books” paper procedure into a Python-style trace.

The facts we will use

Page counts are [80, 150, 90]. Start count=0; for each pages value, add 1 only when pages<100.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-3/23

Block to Python Bridge

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Start counter

count begins at 0 before the loop.



2. Trace decisions

80 qualifies: count 1. 150 does not: count 1. 90 qualifies: count 2.



3. Report count

The final result is 2 qualifying books, not the sum of their page counts.

CHANGE ONE THING

Replace the last page count 90 with 100.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Start counter

What happens and why: count begins at 0 before the loop.

Step: 2. Trace decisions

What happens and why: 80 qualifies: count 1. 150 does not: count 1. 90 qualifies: count 2.

Step: 3. Report count

What happens and why: The final result is 2 qualifying books, not the sum of their page counts.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Read the picture in order. The final result is 2 qualifying books, not the sum of their page counts. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

This deterministic loop counts a stated condition; it neither trains a model nor measures whether the books suit a reader.

Read the labelled picture: Trace the changed list and identify where the result differs. Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

Change one thing: Replace the last page count 90 with 100. Predict the result before checking the explanation. What remains the same?

7. Find and repair a mistake

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak evidence.



COMPARE • DIAGNOSE

Weak idea: Moving blocks into text automatically creates correct Python. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record · Expected · Actual · Likely cause / next test

Evidence record: normal case

Expected: works

Actual: _____

Likely cause / next test: _____

Evidence record: boundary case

Expected: stated rule

Actual: _____

Likely cause / next test: _____

Evidence record: missing input

Expected: safe response

Actual: _____

Likely cause / next test: _____

Evidence record: fresh transfer

Expected: generalizes

Actual: _____

Likely cause / next test: _____

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Save the file with the input values, expected and observed outputs, and a short explanation. This program

follows a written multiplication rule. It has not trained a model, predicted somebody’s preferences or checked that real packs contain the stated number. Its conclusion depends on the supplied data. Week 2 examines the values and types behind these names.

A script runs without an error and displays 11 for five packs of six. Is it correct for the stated requirement? Give the required result, the likely wrong operation and one check after repair. State whether your evidence came from a device run or a paper trace.

8. Practise together

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



SPECIFY THE SYSTEM

Define task, trusted context, constraints, schema, tools, checks, fallback, and stopping rule before trusting output.

SET task, audience, constraints, output_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF evidence missing: ask, retrieve, or abstain RETURN useful output + source trail + limitations

Trace one input. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Change only packs to 5. Predict the result from five groups of six, save, then run the entire file again. The expected output is 30. Keep the original four-pack result beside the new one and label the changed input. A comment can explain an instruction: Python ignores text after # outside a string. A comment does not change the calculation.

```
packs = 5 # five equal packs
per_pack = 6
total = packs * per_pack
print(total)
```

A program can run successfully and still solve the wrong problem. Replacing * with + would display 11 for this input, but addition does not count five groups of six. Use the requirement and a paper calculation to check meaning; the absence of an error message is not proof of correctness.

Practise with less help: Predict then observe

Use workbook W2 for the connected example “From a paper algorithm to Python”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for From a paper algorithm to Python. Proposed explanation: “An equals sign displays the answer”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Trace the changed list and identify where the result differs. Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



PREDICT → RUN → VERIFY → EXPLAIN
Translate five blocks, predict variable values, run or simulate, then test empty-list and missing-field cases.

Translate five blocks, predict variable values, run or simulate, then test empty-list and missing-field cases.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

Test / input: normal

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the worked example.



CREATOR MASTERY
1 Define • 2 Trace • 3 Build • 4 Evaluate • 5 Verify • 6 Defend limits

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or evidence path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.
- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest evidence of independent understanding:

Write or draw your reasoning and evidence.

11. Connect your project

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



VERIFIED STUDY BUDDY

Implement small transparent functions for selection, validation, and citation formatting.

THIS WEEK'S CHECKPOINT Implement small transparent functions for selection, validation, and citation formatting.

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → human decision

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under human control?

Write or draw your reasoning and evidence.

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse fluent output with verified truth.

Evidence for your project

For your Verified Study Buddy, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

12. Show what you know

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



INDEPENDENT CREATOR PROOF

Translate a block workflow into readable Python.

CHALLENGE Translate a block workflow into readable Python.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My evidence that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

SELF-CHECK ☐ I explained the mechanism ☐ I used evidence ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

13. Study a complete case

Week 23 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

A block program sets `total=0` and adds each value in `[2,4,6]`.

The question

Translate and trace.

Worked explanation

Python uses `total=0`, then an indented `total += value` inside for value in `[2,4,6]`. Totals are 2,6,12.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Trace the accumulator inside the loop

Evidence or stage · Value or result

Evidence or stage: Start total

Value or result: 0

Evidence or stage: Add 2

Value or result: 2

Evidence or stage: Add 4

Value or result: 6

Evidence or stage: Add 6

Value or result: 12

A block program sets total=0 and adds each value in [2,4,6].

Translate and trace.

Python uses total=0, then an indented total += value inside for value in [2,4,6]. Totals are 2,6,12.

14. Try a new case independently

Independent case: Changed independent case

Set total=0, then for values 2,4,9 perform total=total+value. Trace the result. What changes if total=0 moves inside the loop before that addition?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 23 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Trace this Python: `score = 2; then if score >= 2: score = score + 3`. What is the final value, and what changes if the condition becomes `score > 2`?

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Implement small transparent functions for selection, validation, and citation formatting.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

Write Python that totals [3,5,7], then trace each total. Explain how indenting 'total=0' inside the loop changes the result.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

15. Your lesson route

Start with this question: Translate a “count qualifying books” paper procedure into a Python-style trace.

By the end, I can · How I will show it

By the end, I can: Translate a short paper calculation into a Python script

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Predict and explain the order of executed statements

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Keep expected and observed output separate when checking a program

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

16. Finish and return

Close your notes. Explain this goal in your own words: Translate a short paper calculation into a Python script

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.